Copernicus Marine Service

QUALITY INFORMATION DOCUMENT

WAVE Thematic Assembly Centre:

WAVE_GLO_PHY_SPC_L3_NRT_014_009

Issue: 1.0

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Approval date by the CMEMS product quality coordination team: 28/05/2024

CHANGE RECORD

When the quality of the products changes, the QuID is updated and a row is added to this table. The third column specifies which sections or sub-sections have been updated. The fourth column should mention the version of the product to which the change applies.

Issue	Date	§	Description of Change	Author	Validated By
1.0	May 2024	All	Creation of the document	C. Peureux, A. Nigou, A. Ollivier	P. Zunino, R. Husson

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I EXECUTIVE SUMMARY

I.1 Products covered by this document

The WAVE_GLO_PHY_SPC_L3_NRT_009 CFOSAT 2D spectral product is an innovative product, providing a characterization of sea states. Its assessment is presented and assessed by comparison with WAM model and Sentinel-1 data.

This document describes the estimation process and the quality of the operational (NRT) spectral products listed hereafter:

Product	WAVE_GLO_PHY_SPC_L3_SPC_NRT_009
Area	Global ocean
Missions	CFOSAT
Main information	- 2D spectral data - Partition mask - Parameters integrated from the 2D spectral data - Parameters integrated from the 2D spectrum after partitioning of the sea states
Spatial resolution	70 km along track with left and right positioning, 45 km apart from the nadir ground track
Temporal resolution	3-hour

Table 1: Presentation of product's main information with its spatial and temporal resolution.

	Temporal availability	
	Begin date	**End date**
CFOSAT	25 Avril 2019	Present

Table 2: Temporal period processed by the L3 SPC wave chain.

I.2 Summary of the results

The performance of the SWIM instrument from CFOSAT's was extensively described in the literature ([1], [2], [3], [4]). This innovative mission already exceeded the mission's specifications and keeps on improving thanks to regular ground segment evolutions and reprocessing.

Due to its instrument design, CFOSAT is complementary to the existing remote sensing missions to characterize waves. Whereas altimetry estimated the total energy of waves (Hs), CFOSAT, like Sentinel-1 wave modes (see [8], [9] and [10]), can determine how the energy is distributed in direction and wavelength.

Unlike Sentinel-1 which essentially catches long period swells (due to a cutoff effect described in [8], CFOSAT is designed to observe all types of swells from 30 m to 500 m (and soon up to 1200 m) wavelength without any cut off.

SWIM resolution and coverage are also complementary to Sentinel-1's. CFOSAT has a regular sampling on all oceans up to 82°N/S every 70 km, whereas sentinel-1 wave mode is not available in Northern Atlantic and has a 100 km sampling with incidence angle alternated between 23° and 26°.

On the other hand, users have to be aware of a few limitations for SWIM data.

- Parasitic peaks affecting young seas (and so typically closed seas) and enhanced for height spectra compared to slope spectra (see below) -
- Non availability of wavelengths over 500 m (in L2, but future version should be relaxed to 1200 m)

Important information These products contain the full spectra at the original sampling rate. They are complementary to fireworks SWIM products [10] which content only the swell part, propagated in other time and space of the satellite passage.

I.3 Estimated Accuracy Numbers

Table 3: Summary of the estimated accuracy numbers of SWIM product compared to the collocated MFWAM ocean wave model (for SWIM cycle 143, see more details in Section IV.2.2).

Variables	Units	Metrics	Wave classification	Bias	RMSD	SI	Correlation
Hs (VAVH)	m	Mean (inside the 70/90km observed box)	Swell 1	-0.20 m	0.68 m	33.48%	82.45%
			Swell 2	0.14 m	0.71 m	54.02%	47.39%
			Wind waves	-0.25 m	0.76 m	31.38%	88.64%
Wavelength (extracted from period VTPK)	m	Mean	Swell 1	-1.23 m	68.12 m	36.20%	67.41%
			Swell 2	-13.85 m	87.50 m	41.09%	70.00%
			Wind waves	12.31 m	22.47 m	24.12%	89.57%
Direction (VPED)	Degree	Peak (maximal value)	Swell 1	-0.41°	30.97°	30.53%	86.61%
			Swell 2	-1.01°	41.42°	38.95%	81.48%
			Wind waves	-0.17°	37.55°	34.23%	81.91%

II PRODUCTION SYSTEM DESCRIPTION

II.1 Production center name

WAVE-CLS-TOULOUSE-FR

II.2 Operational system name

The operational chain is given a generic name: the L3 SPC wave chain.

II.3 Production centre description for the version covered by this document

II.3.1 Acquisition

The spectral measurements used in the system are Level-2P products from CFOSAT/SWIM instrument. Their source, delay and period of availability are summarised in Table 4.

L3-wave products availability period will be the same as spectral data from L2 (see Table 4). Note that it is shorter than L2 nadir due to anomalies identified at the beginning of the mission on spectral data only between October 2018 and April 2019.

Since their production started, L2 and L2P have been generated using several successive versions of the CNES CFOSAT/SWIM processor. Thus, a recent L2 and L2P reprocessing was performed with the latest version of the processor to generate a homogeneous L3 dataset since the beginning of L2P availability period.

Mission	Type of product	Source	Availability delay	Period of availability
CFOSAT	L2/L2P/L3	CLS/CNES TREX	< 3hours	25th April 2019 - present

Table 4: Source, delay and period of availability of the data.

Mission	Cycle duration (days)	Number of orbits per cycle	Sun-synchronous	Acquisition mode
CFOSAT	13	197	Yes	10° incidence angle

Table 5: SPC mission characteristics.

The purpose of the acquisition processing is to acquire all the files necessary to compute the L3 products for each data source. This includes the L2p files.

II.3.2 L3 generation

The main input files for this processing are the SWIM Level-2 products provided by the CNES CWWIC Centre and made available on the Aviso website [3].

Wave spectra model output is given as wave height spectra, between 0 and 360°. Due to SWIM instrument limitations (signal-to-noise ratio for small wave number and directional ambiguity), note that:

- SWIM wave spectra are expressed as wave slope spectra, because of the enhanced noise level of wave height spectra.
- In Level-2 products, spectra is output between 0 and 180°. To make SWIM data directly comparable with model data post-processing is carried out. The spectrum is duplicated over the missing half of full azimuthal range to obtain wave spectra expressed between 0 and 360° (obtaining L2P/L3 products).

All the processes for switching from Level-2 to L2P products are listed in [5] including:

- data editing;
- processing of the SWIM slope spectrum (central symmetry process and filtering of parasitic peaks);
- partitioning;
- estimation of wave integrated parameters.

The procedure used to generate L3 products from L2P essentially involves renaming and restricting the number of variables in the product, in order to be more user-friendly and to be in line with Copernicus Marine Service requirements.

As mentioned in the L2P user handbook [5] the spectra in the L2 products are provided between 0° and 180°, because of the directional ambiguity inherent to the SWIM measurements. In the L2PBOX, the spectra are given in the 0°to 360° range, obtained by performing a central symmetry. In other words, the spectrum is duplicated over the missing half of full azimuthal range. Since this procedure doubles the total energy present in the spectrum, each value of the 2D spectrum is divided by 2, to conserve the total energy.

III VALIDATION FRAMEWORK

The validation aims to control the quality of the produced wave parameters (significant wave height, wavelength, direction) based on SWIM full spectra and its partitions . The performance is assessed by cyclic reports available on Aviso web site [11] concerning:

- The data availability and spatial/temporal coverage;
- The quality flags;
- Comparison to nadir data;
- Comparison to Sentinel-1 and WAM outputs.

In this document, we explore the added value of filtering parasitic peaks and we added scatter plots of comparison with WAM MétéoFrance model [12], colocalised at each CFOSAT ground track.

IV VALIDATION RESULTS

IV.1 SWIM 2D spectral matrix

The SWIM 2D spectral data are matrix depending on:

- **A wave number, related to the wave period (the ray of the slope spectrum)**
- **A direction (related to the wave direction of propagation (with a 180° ambiguity inherent to the SWIM measurement)**

They are available at a 70 km resolution along track, at 45 km apart from the nadir observation as shown on the plot below:

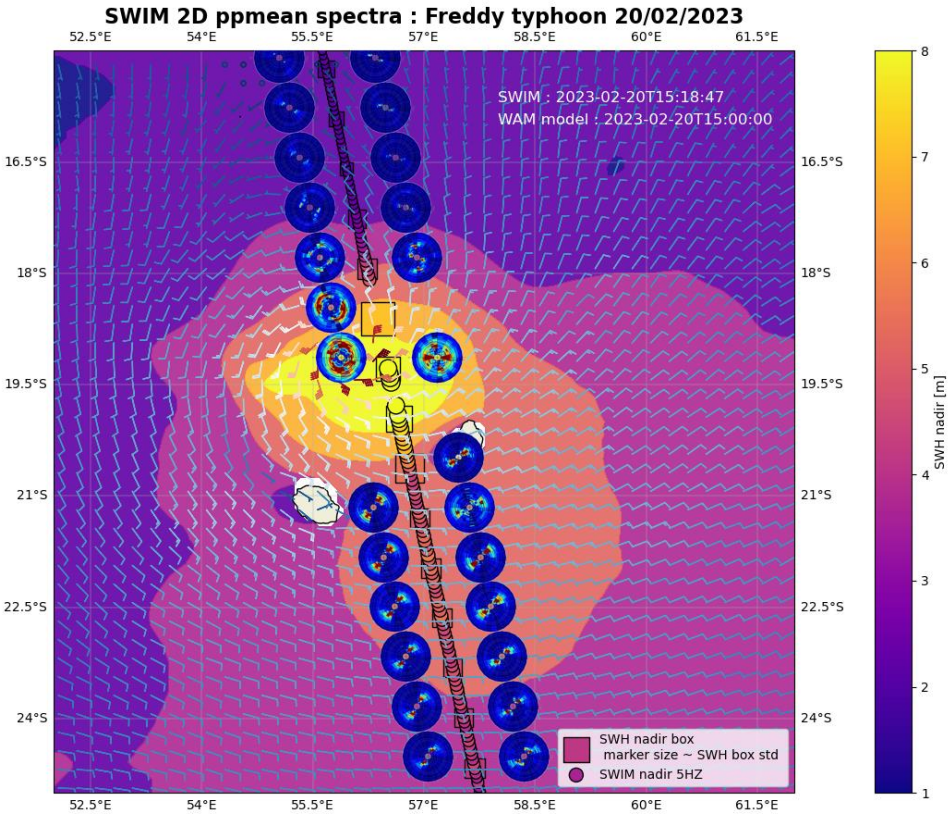

Figure 1: Along track view of the spectral product during Freddy typhoon. 70 km resolution at 45 km apart, on left and right side, of the nadir track.

IV.2 Quality assessment, limitation

The performance of the SWIM instrument from CFOSAT's was extensively described in the literature ([1], [2], [3], [4], [6]). This innovative mission already exceeded the mission's specifications and keeps on improving thanks to regular ground segment evolutions and reprocessing.

Due to its instrument design, CFOSAT is complementary to the existing remote sensing missions to characterise waves (see Figure 2 & 4). Whereas altimetry estimated the total energy of waves (Hs), CFOSAT, like Sentinel-1 wave modes (see [8], [9] and [10]), can determine how the energy is distributed in direction and wavelength. But unlike Sentinel-1 which essentially catches long period swells (due to a

cutoff effect described in [8], CFOSAT is designed to observe all types of swells from 30 m to 500 m (and soon up to 1200 m) without any cut off.

When averaged over a 10°/10° box, the central symmetry process (as described in Section II.3.2) is carried out on spectra from SWIM, WAM, and S1 and highlight that the cut off does not affect SWIM (see Figure 3).

SWIM resolution and coverage is also complementary to Sentinel-1's, as shown on (Figure 2). CFOSAT has a regular sampling on all oceans up to 82°N/S every 70 km, whereas sentinel-1 wave mode is not available in Northern Atlantic and has a 100 km sampling with incidence angle alternated between 23° and 26°.

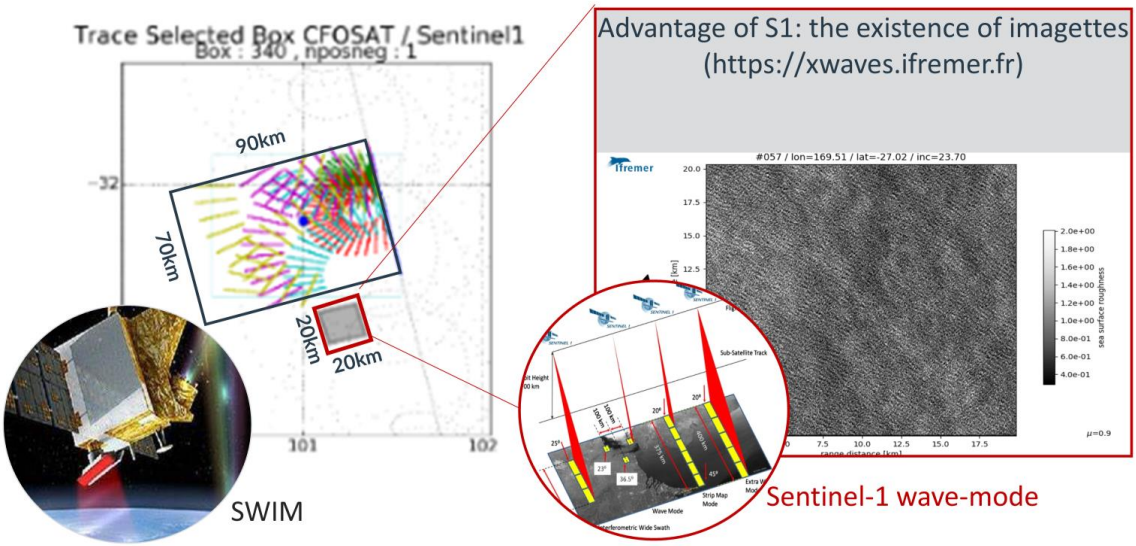

Figure 2: SWIM and Sentinel-1 resolution and ground track collocated.

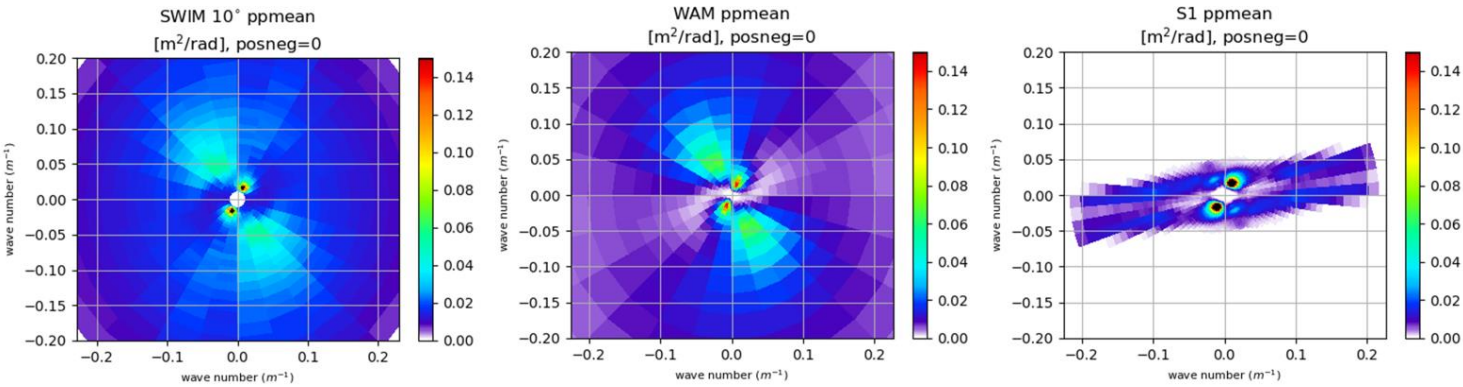

Figure 3: Mixed Sea example for SWIM, symmetrised WAM and symmetrised Sentinel1 averaged over 10°/10° box. From C. Peureux in Ollivier et al.'s talk OSTST 2022 Venice.

On the other hand, users should be aware of a few limitations for SWIM data.

- Parasitic peaks affecting young seas (and so typically closed seas) and enhanced for hight spectra compared to slope spectra (see below).
- Non availability of wavelengths over 500 m (in L2, but future version should be relaxed to 1200 m).

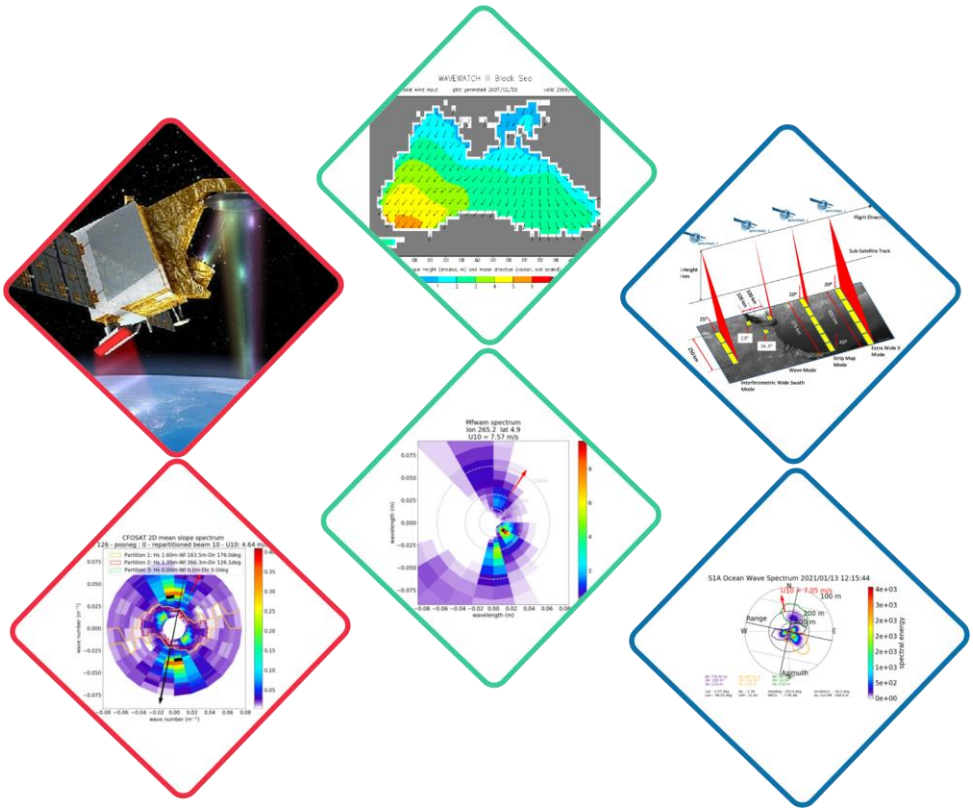

Figure 4: Complementarity of different observation systems looking at the same scene: SWIM, MFWAM model and Sentinel-1. From C. Peureux in Ollivier et al.'s talk OSTST 2022 Venice.

IV.2.1 Integrated parameters data (Hs, direction, wavelength)

The wave system parameters (significant wave height, peak wavelength, and peak direction) are calculated over the whole symmetrized slope spectrum of the 10° beam. The peak wavelength and peak direction correspond to the dominant wavelength and direction in the 2D wave slope spectrum.

Scatter plots of SWIM integrated wave parameters compared with collocated MFWAM data are represented in Figure 5. MFWAM assimilates different products (SWIM L2) that is why results are not rigorously independent.

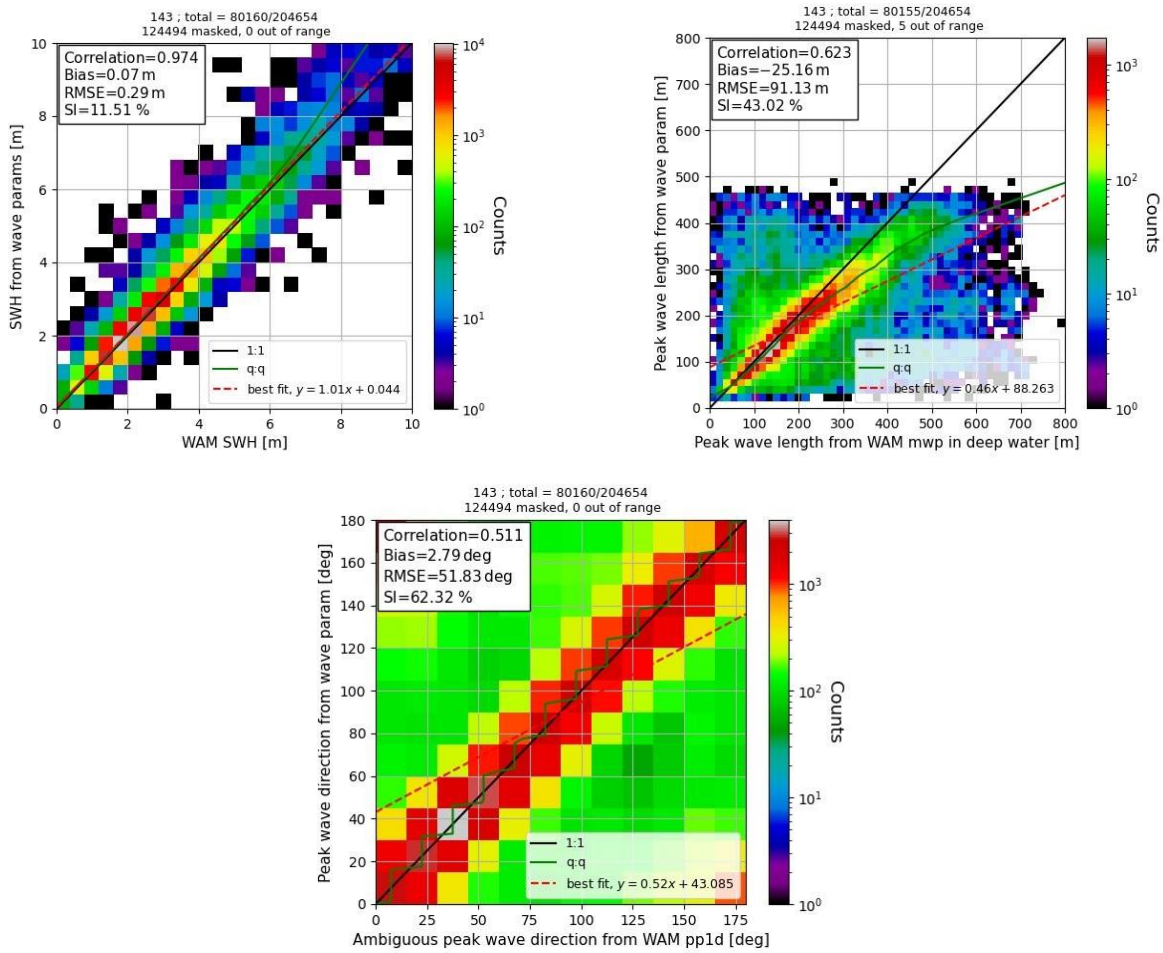

Figure 5: scatter plots of SWIM vs collocated MFWAM ocean wave model output integrated parameters (significant wave height, peak wave number and peak direction) for SWIM cycle 143.

Statistical indicators for SWIM integrated wave parameters compared with MFWAM model are summarized in Table 6 (results obtained over 1 cycle).

Cycle 143	**Bias**	**RMSD**	**SI**	**Correlation**
Hs	0.07 m	0.29 m	11.51%	97.40 %
Wavelength	-25.16 m	91.13 m	43.02%	62.30%
Direction	2.79°	51.83°	62.32%	51.1%

Table 6: statistical indicators of SWIM vs collocated MFWAM ocean wave model output integrated parameters (significant wave height, peak wave number and peak direction) for SWIM cycle 143.

Figure 6 highlights a good agreement between estimated wave parameters from SWIM 2D slope spectra and the associated wave parameters from MFWAM collocated height spectra.

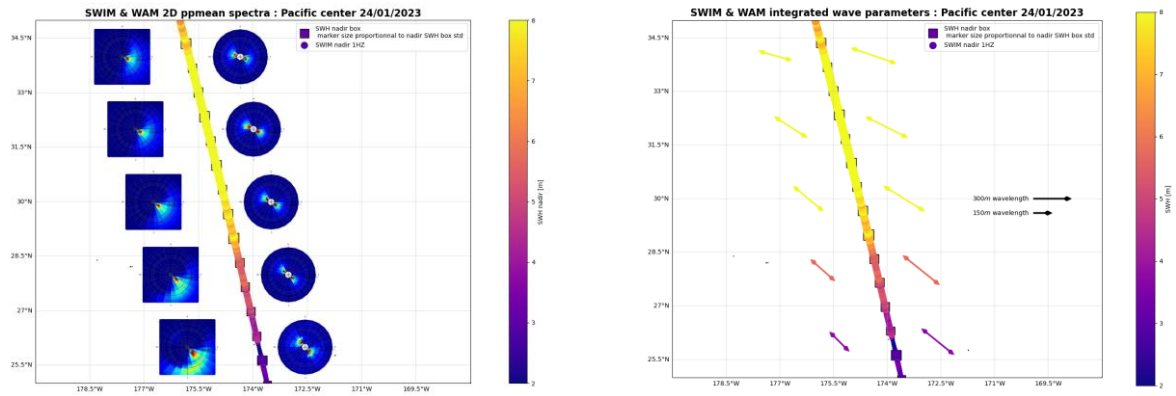

Figure 6: Along track representation of SWIM slope spectra and WAM height spectra every 2° of latitude (on the left) and the associated integrated wave parameters represented by arrows (colour: SWH, length: peak wavelength, direction: peak direction)

IV.2.2 Integrated parameters data (Hs, direction, wavelength) from partition spectra

The entire slope spectrum of the 10° beam is partitioned into three wave systems detectable by SWIM (see Figure 7). The detailed procedure of partitioning can be found in the L2P user handbook [5]. For each of these partitions, we get an estimation of the wave system parameters (SWH, peak wavelength, and peak direction).

The partitions are ranked by decreasing significant wave height.

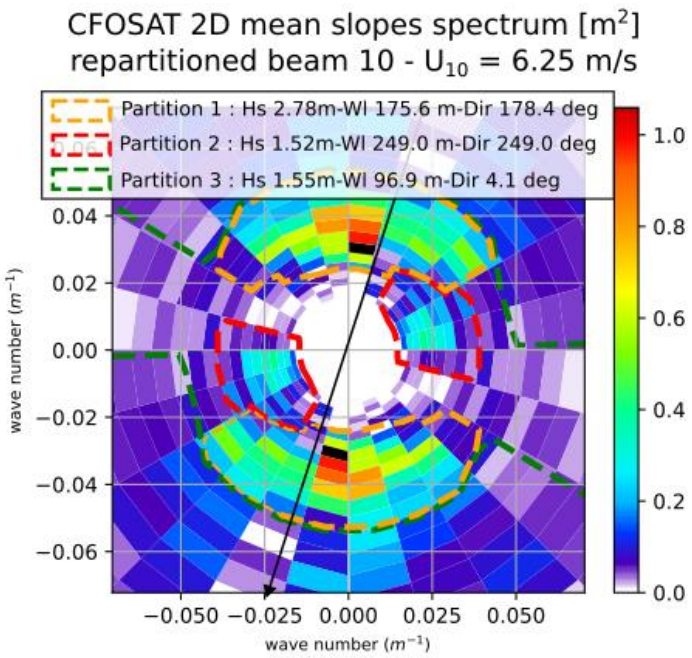

Figure 7: SWIM 2D slope spectrum partitioned. Each partition is represented by a different colour. Integrated parameters of each partition are indicated in the legend.

Figure 8 highlights differences between estimated wave parameters from each partition of SWIM 2D spectra and wave parameters estimated from partitioned MFWAM collocated height spectra.

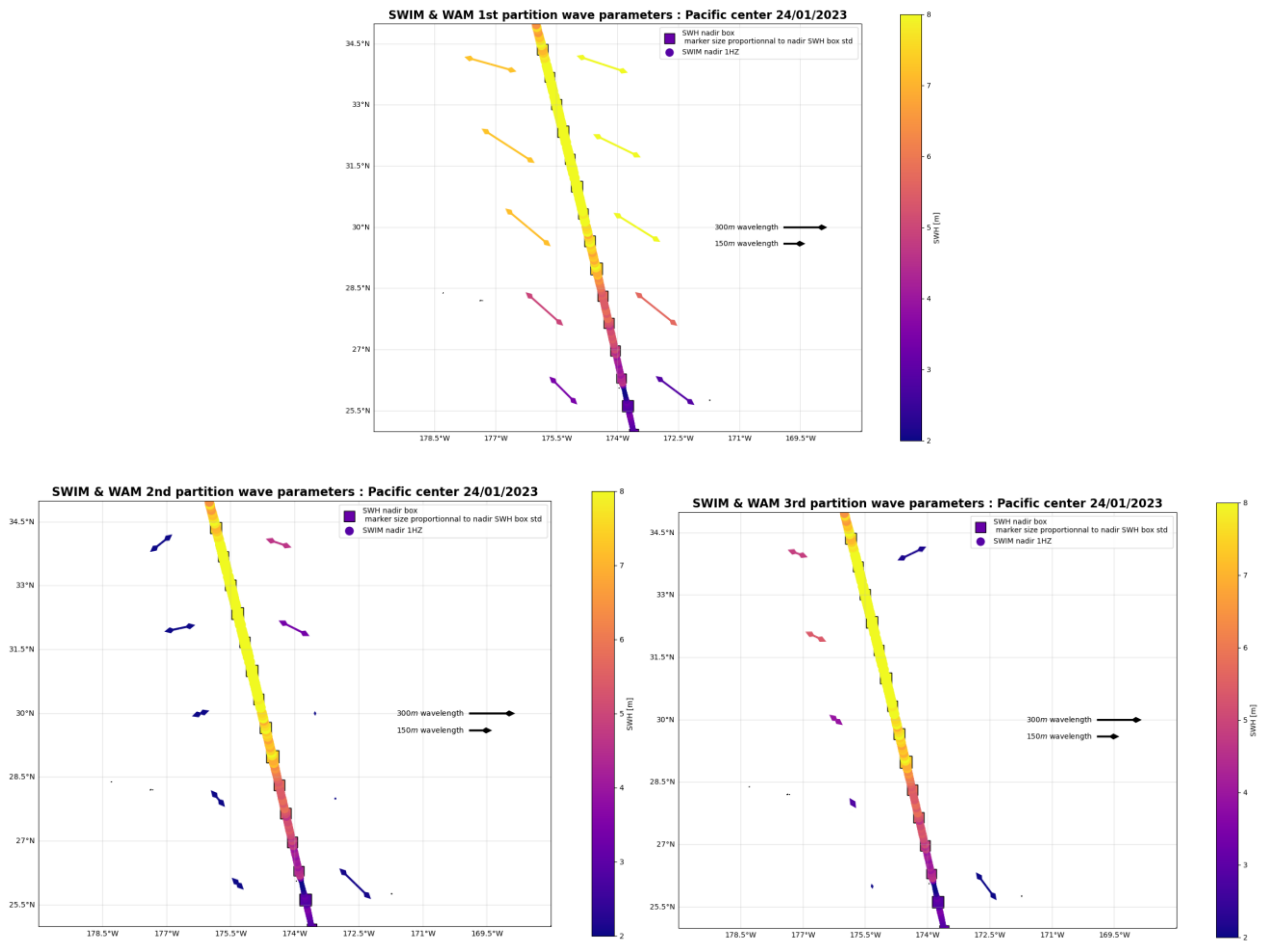

Figure 8: Along track representation of integrated wave parameters from SWIM (on right of the track) & MFWAM model (on the left) for each spectral partition. Wave parameters are represented by arrows (colour: SWH, length: peak wavelength, direction: peak direction)

Even if the first partition of SWIM (Figure 8) is in line with the first swell partition of MFWAM, second and third partitions seen by SWIM are not in good agreement with the second swell partition and wind waves partition of MFWAM model.

Indeed, unlike SWIM partitioning, wave model partitioning such as MFWAM associates each partition with a sea state (in this case main swell, secondary swell and wind sea). It is therefore necessary, in order to make relevant comparison, to associate SWIM partitions with WAM partitions, and thus with a corresponding sea state. To do this, a cross-assignment method is used that associates each SWIM partition with the closest WAM partition (via a spectral distance minimization process). This enables us to compare SWIM and WAM partitions by sea state separation.

Plots in Figure 9 (a, b & c) represent the comparison of integrated wave parameters between SWIM and WAM collocated model for each partition (swell1, swell2 and wind waves) after making a cross assignment over one CFOSAT cycle.

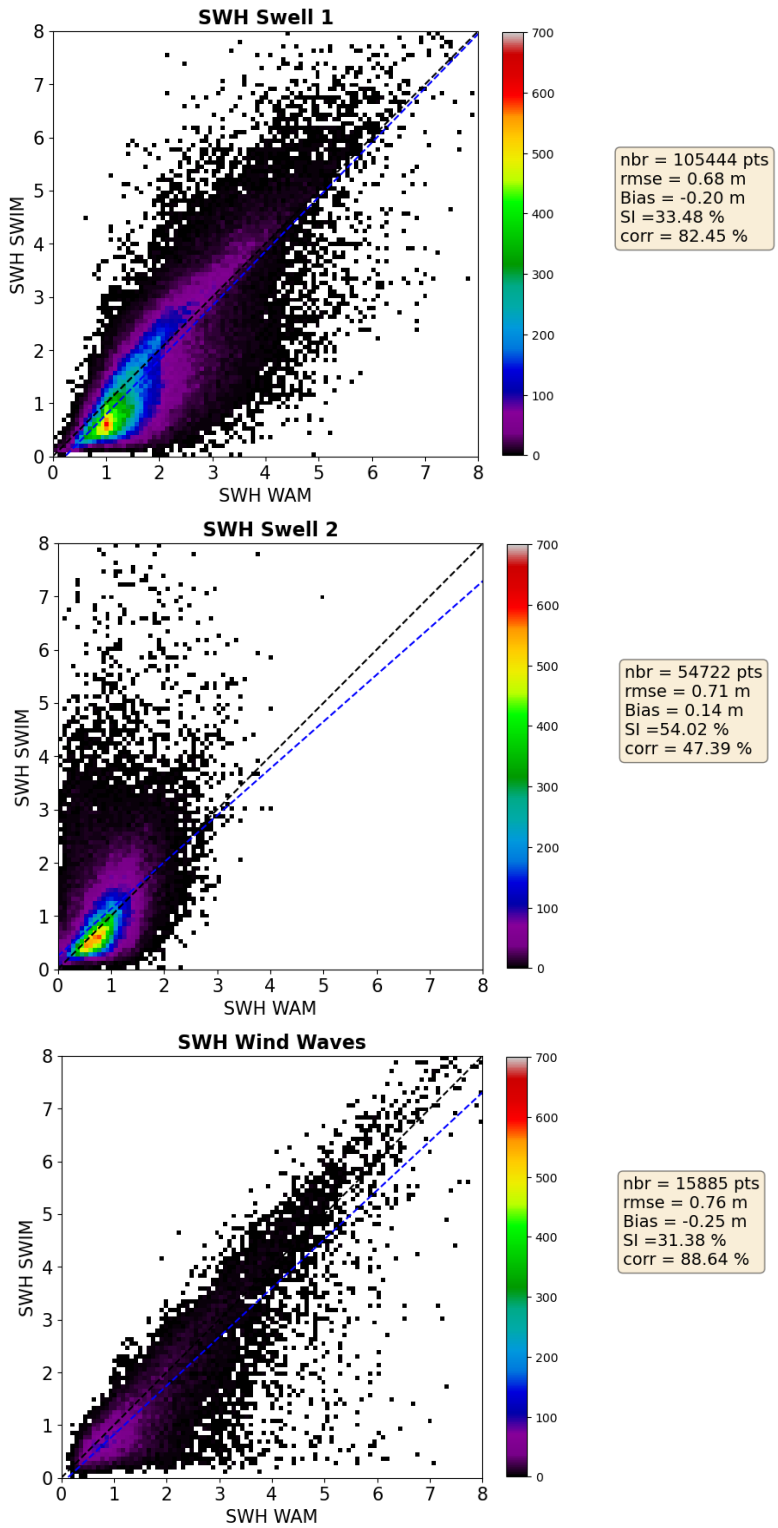

Figure 9 (a): Scatter plot of partitioned wave parameters (SWH) from SWIM & MFWAM after making a cross assignment over CFOSAT cycle 143.

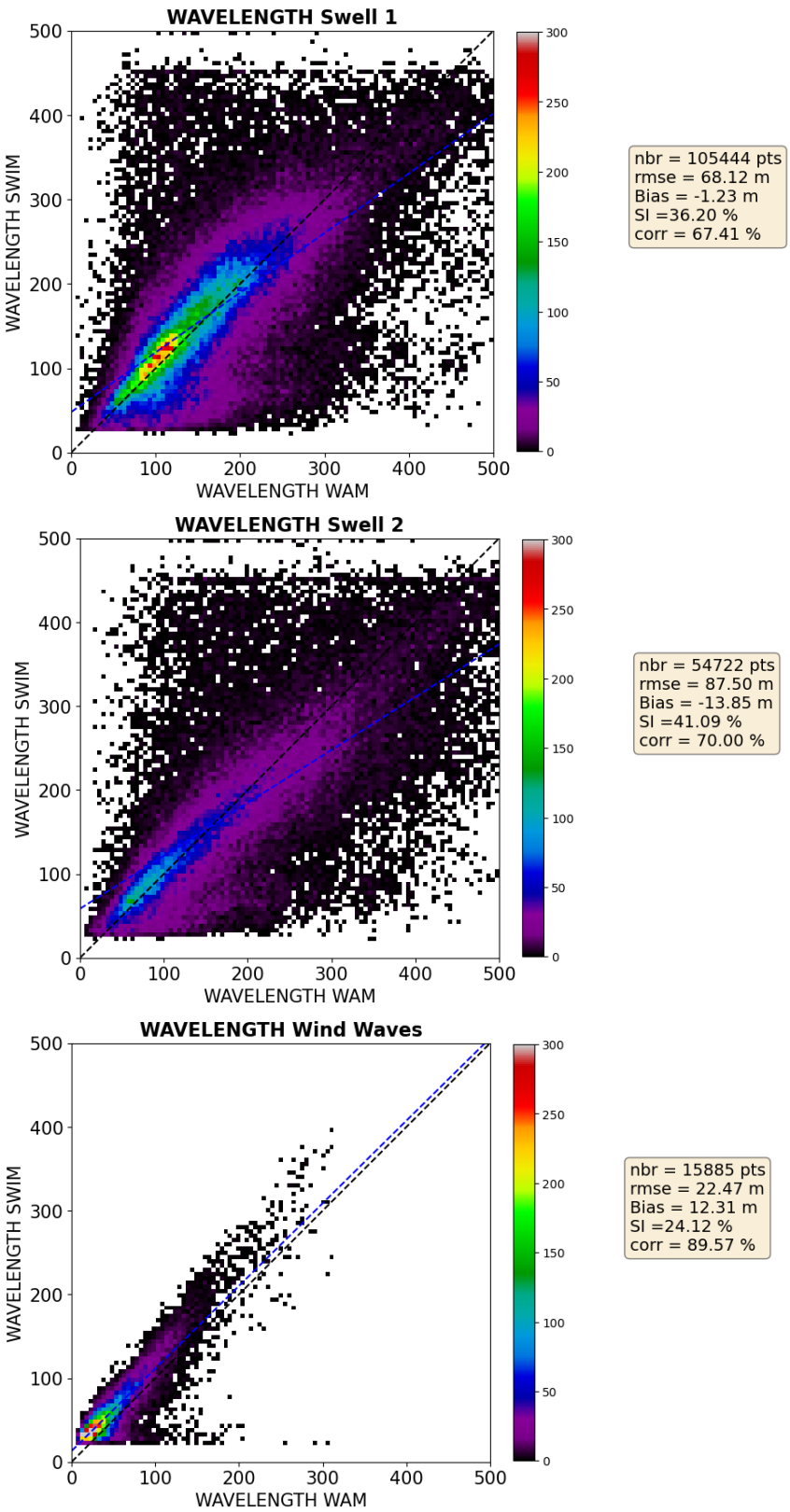

Figure 9 (b): Scatter plot of partitioned wave parameters (WAVE LENGTH) from SWIM & MFWAM after making a cross assignment over CFOSAT cycle 143.

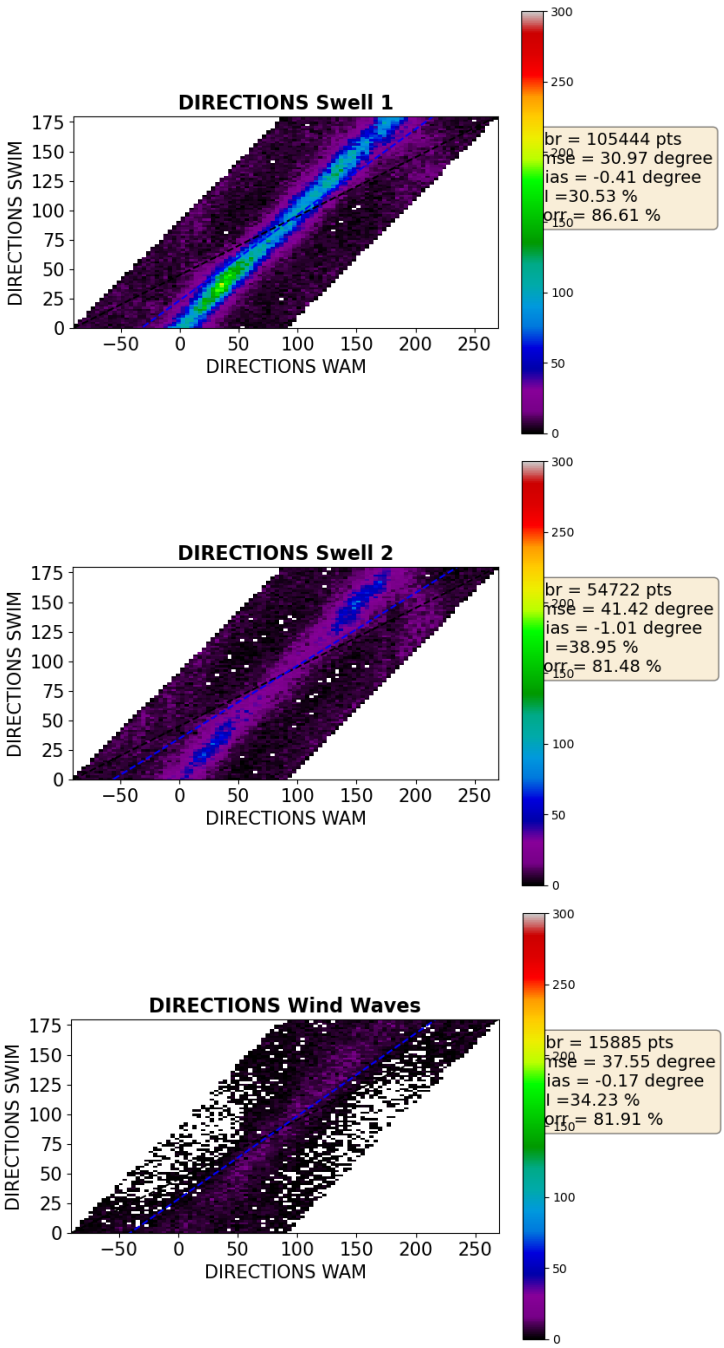

Figure 9 (c): Scatter plot of partitioned wave parameters (DIRECTION) from SWIM & MFWAM after making a cross assignment over CFOSAT cycle 143.

Statistical indicators for SWIM partitioned integrated wave parameters compared with MFWAM model are summarized Table 7.

Cycle 143		**Bias**	**RMSD**	**SI**	**Correlation**
Hs	Swell 1	-0.20 m	0.68 m	33.48%	82.45%
	Swell 2	0.14 m	0.71 m	54.02%	47.39%
	Wind waves	-0.25 m	0.76 m	31.38%	88.64%
Wavelength	Swell 1	-1.23 m	68.12 m	36.20%	67.41%
	Swell 2	-13.85 m	87.50 m	41.09%	70.00%
	Wind waves	12.31 m	22.47 m	24.12%	89.57%
Direction	Swell 1	-0.41°	30.97°	30.53%	86.61%
	Swell 2	-1.01°	41.42°	38.95%	81.48%
	Wind waves	-0.17°	37.55°	34.23%	81.91%

Table 7: statistical indicators of SWIM vs collocated MFWAM ocean wave model output partitioned integrated parameters (significant wave height, peak wave number and peak direction) for SWIM cycle 143.

V SYSTEM'S NOTICEABLE EVENTS, OUTAGES OR CHANGES

This section is dedicated to track and describe changes that may occur in the current operational system due to outages, upstream data changes (e.g. addition or loss of a satellite, inclusion of other in-situ data).

Date	Change/Event description	System version	other
	Not applicable		

VI QUALITY CHANGES SINCE PREVIOUS VERSION

Since the demonstration products delivered to internal users and discussed during 2023 TWAPAS meeting, the parasitic pics were filtered.

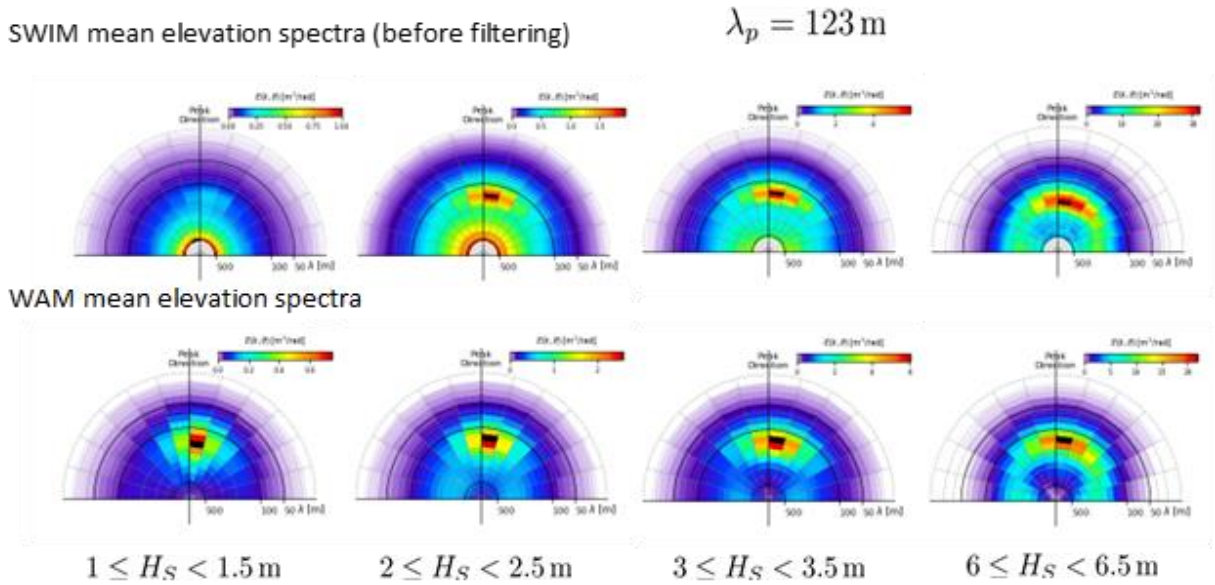

Figure 10: Mean 2D elevation spectra with a peak wavelength of 123 m for various significant wave heights for SWIM (before parasitic peaks filtering) and collocated MFWAM ocean wave model outputs for the year 2021. Parasitic peaks appear on the lowest significant wave height spectra with an energetic circle around 500 m length.

Indeed, a difficulty with SWIM measurements is dealing with the so-called parasitic peaks [7]. They are observed in young sea conditions, with significant wave heights typically lower than 2.5 m and short dominant wave periods. Although their origin is still under investigation, they are now well characterized. Parasitic peaks arise as quasi-random large-scale fluctuations in wave spectra, at wavelength typically larger than 300 m (see Figure 10). 2D elevation spectra, E, are obtained from 2D slopes spectra, F, by:

$$E(k,\varphi) = \frac{F(k,\varphi)}{k^2}$$

where k is the wave number, and φ the waves propagation direction. For these reasons, parasitic peaks are enhanced by moving from slopes to elevations spectra. Elevation spectra are more commonly used than slopes spectra, especially for users used to ocean wave model outputs. Parasitic peaks are also particularly prominent on 1D elevation spectra (see Figure 11).

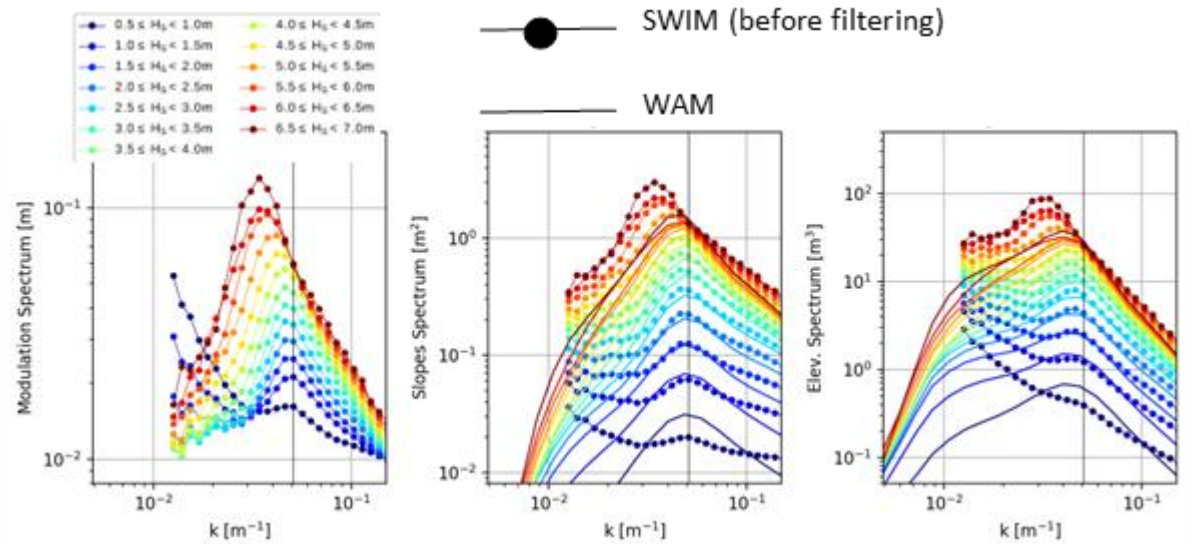

Figure 11: Mean 1D modulation, slopes and elevation spectra with a peak wavelength of 123 m for various significant wave heights for SWIM (before parasitic peaks filtering) and collocated MFWAM ocean wave model output for the year 2021. Parasitic peaks are the cause of the discrepancy between SWIM and MFWAM spectra at low wave numbers, particularly towards low significant wave heights.

The method used in this dataset consists of filtering out spurious peaks by exploiting their random nature. To do this, a signal-to-noise ratio is estimated around each pixel of a SWIM slope spectrum as follows:

$$SNR(k,\varphi) = \frac{\langle F(k,\varphi)\rangle}{\sqrt{\langle F(k,\varphi)^2\rangle - \langle F(k,\varphi)\rangle^2}}$$

where the average $\langle \dots \rangle$ relates to a 3 by 3 square pixels matrix centered on the current (k,φ) pixel. Parasitic peaks are assumed to be associated with low signal-to-noise ratios. The filtering is performed by filtering out these pixels: pixels with signal-to-noise ratio below 1.1 are affected a 0 value. In order to conserve the whole energy, the energy filtered out is reaffected to the whole spectrum so that the total spectrum variance before and after filtering remain equal. After parasitic peaks filtering, the spectrum quality is enhanced (see Figure 12), especially in enclosed basins such as the Gulf of Mexico and the Mediterranean Sea.

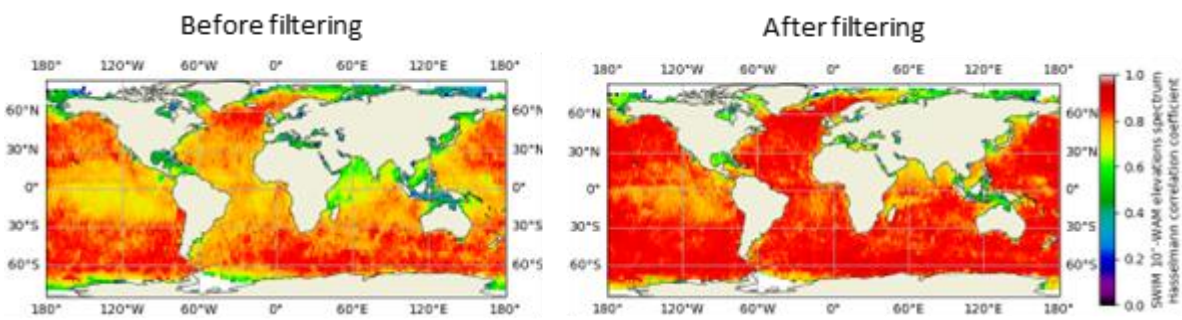

Figure 12: Average correlation coefficient between SWIM and collocated MFWAM ocean wave model outputs spectra for the year 2021, before and after parasitic peaks filtering.

Significant improvements in the estimation of integrated wave parameters after parasitic peak filtering are highlighted in Figures 13 & 14 (a, b, c).

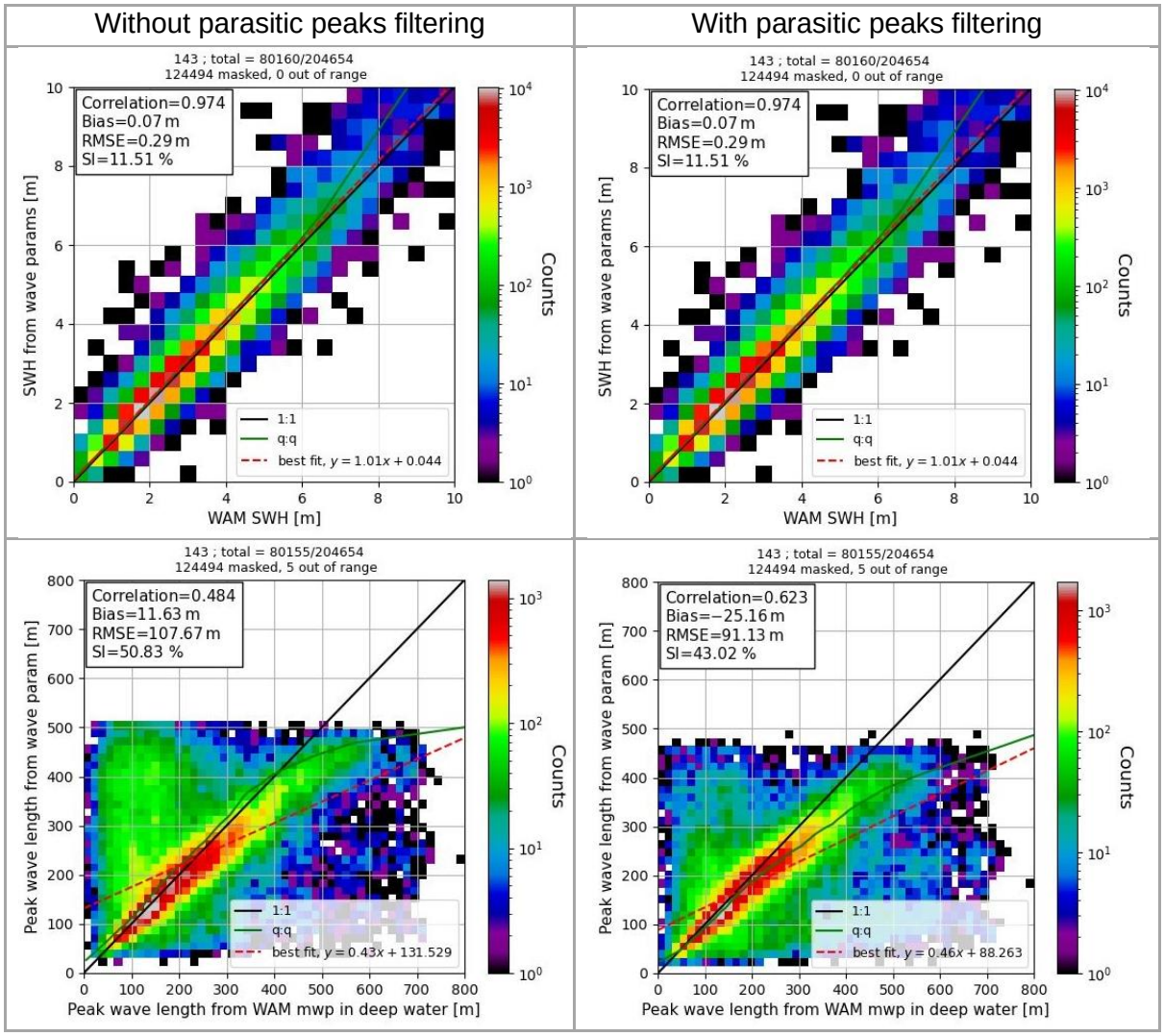

Figure 13: Scatter plots of SWIM vs collocated MFWAM ocean wave model output integrated parameters (significant wave height, peak wave number and peak direction) for SWIM cycle 143, without parasitic peaks filtering (left column) and with parasitic peaks filtering (right column) (continues in next page).

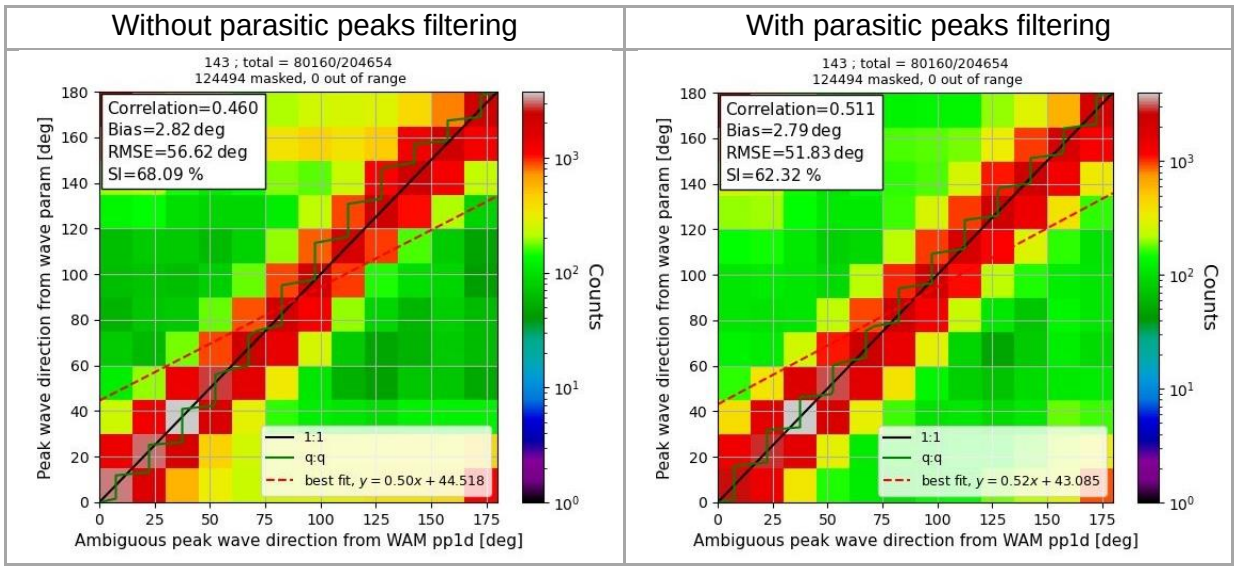

Figure 13: (continued) Scatter plots of SWIM vs collocated MFWAM ocean wave model output integrated parameters (significant wave height, peak wave number and peak direction) for SWIM cycle 143, without parasitic peaks filtering (left column) and with parasitic peaks filtering (right column).

Same enhancements can be observed when looking at statistical indicators from comparison of partitioned integrated wave parameters from SWIM and collocated MFWAM wave model after cross assignment (Figure 14 a, b, c).

On going evolutions based on users' feedback are still on L2 ground segment side (extension of 500 m to 1200 m wavelength), noise reduction.

Without parasitic peaks filtering **With parasitic peaks filtering**

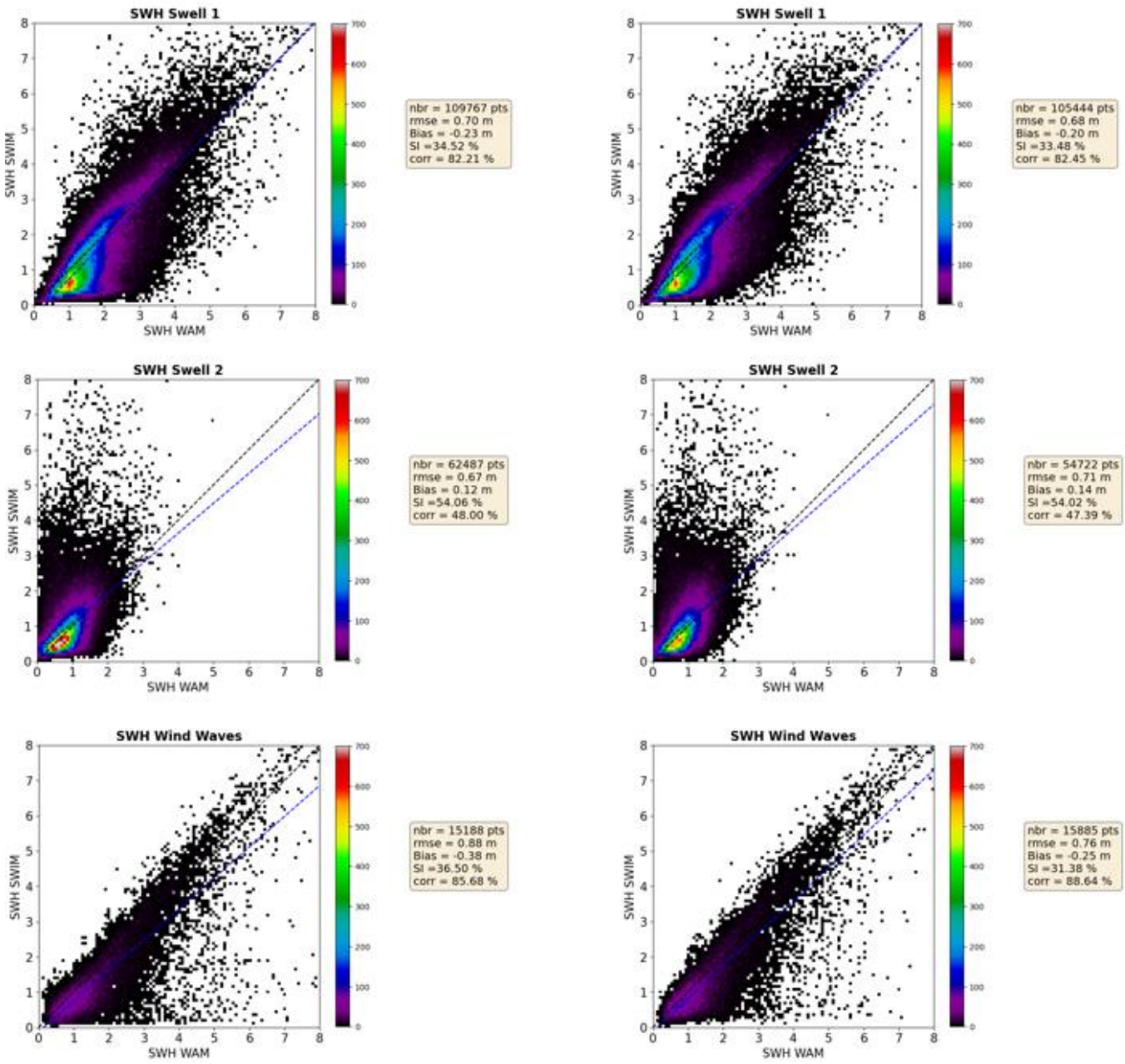

Figure 14 (a): Scatter plot of partitioned wave parameters from SWIM & MFWAM (SWH) after making a cross assignment over CFOSAT cycle 143 – Comparison with/without filtering parasitic peaks.

Without parasitic peaks filtering | **With parasitic peaks filtering**

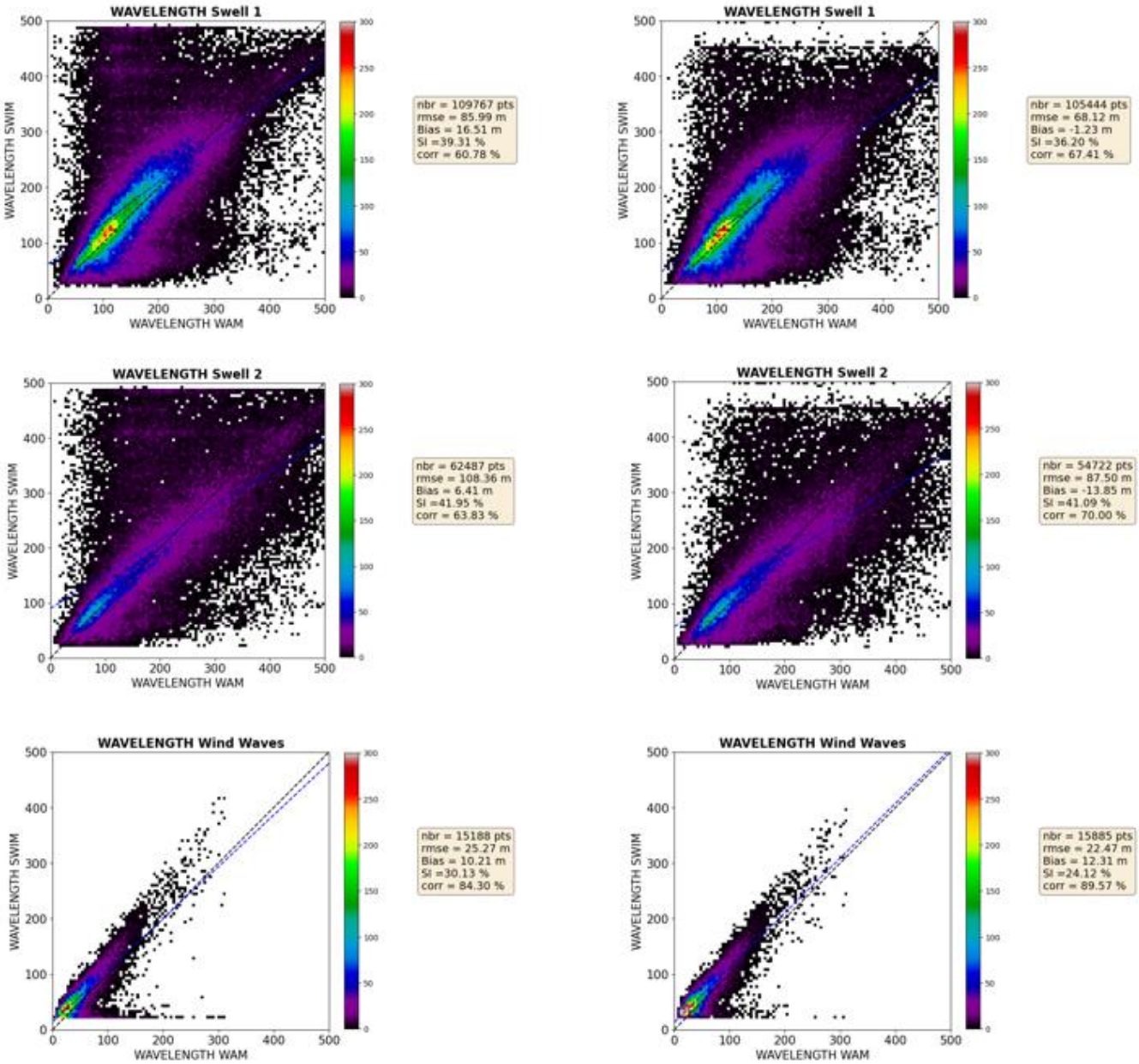

Figure 14 (b): Scatter plot of partitioned wave parameters from SWIM & MFWAM (wavelength) after making a cross assignment over CFOSAT cycle 143 – Comparison with/without filtering parasitic peaks.

Without parasitic peaks filtering **With parasitic peaks filtering**

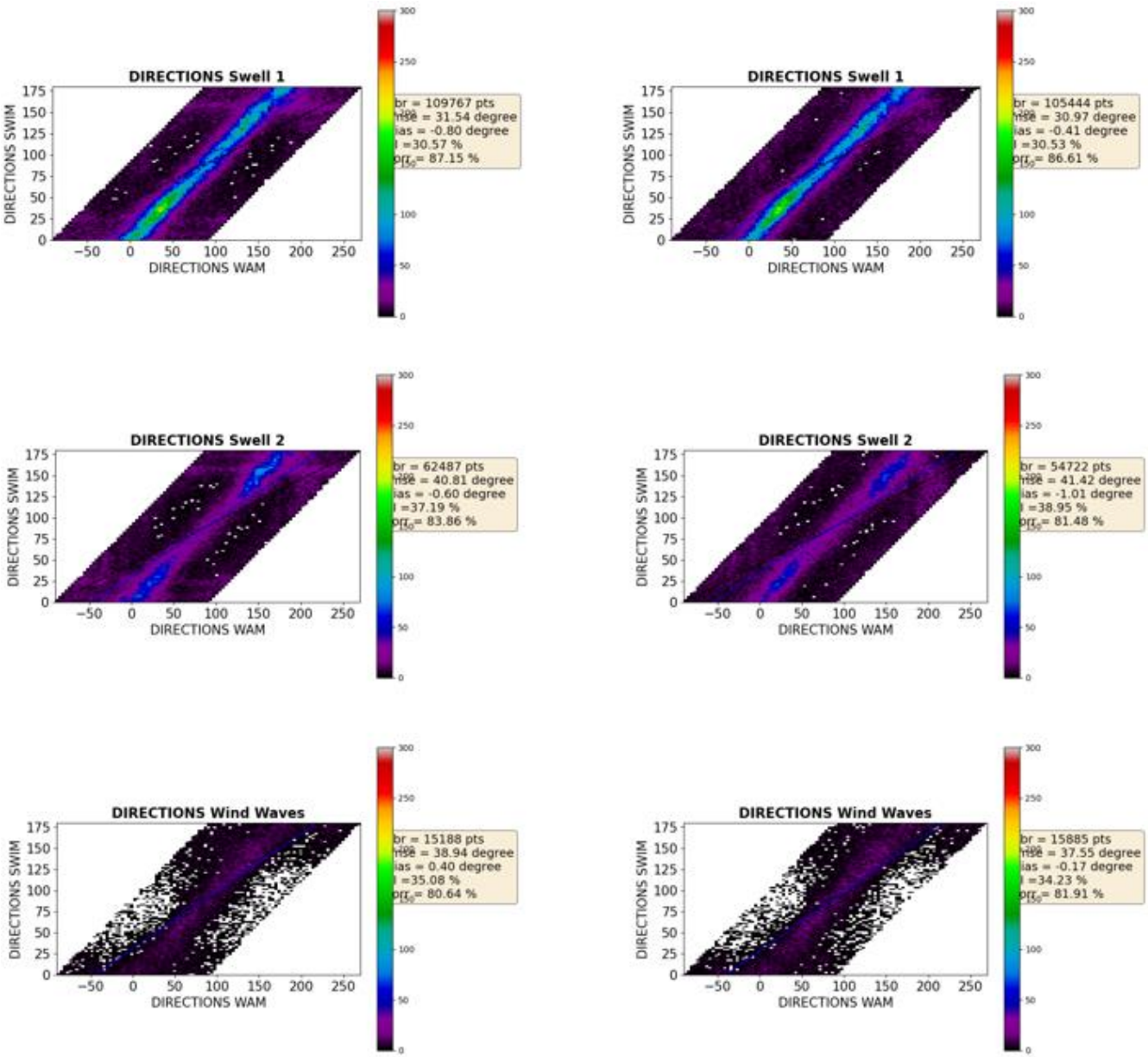

Figure 14 (c): Scatter plot of partitioned wave parameters from SWIM & MFWAM (direction) after making a cross assignment over CFOSAT cycle 143 – Comparison with/without filtering parasitic peaks.

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